

2022 HKDSE Physics

Paper 1A — Multiple-choice Questions

Original questions and official answers with detailed English reasoning. Calculations show the governing relationships, substitution and units; each structured subpart is presented separately.

Editable questions with official answers and detailed English reasoning. Subscripts and superscripts remain editable.

Questions

Official answers are placed directly below each question.

Question 1

A well-insulated vessel with negligible heat capacity contains warm water at $45\text{ }^{\circ}\text{C}$. The temperature of the warm water drops by $5\text{ }^{\circ}\text{C}$ after mixing with 50 g water at temperature $0\text{ }^{\circ}\text{C}$ added to the vessel. Find the original mass of warm water in the vessel.

- A. 400 g
- B. 450 g
- C. 500 g
- D. 550 g

Official answer: A

The final temperature is $40\text{ }^{\circ}\text{C}$. With no heat lost to the surroundings, $mc(45-40) = 50c(40-0)$, using masses in grams. Thus $m = 400\text{ g}$. Answer: A.

Question 2

When ice melts into water at $0\text{ }^{\circ}\text{C}$, what happens to the molecules during the melting process?

- (1) Their average separation increases.
 - (2) Their average kinetic energy increases.
 - (3) Their average potential energy increases.
- A. (1) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (2) and (3) only

Official answer: B

Melting at constant temperature leaves the average molecular kinetic energy unchanged. The supplied latent heat raises molecular potential energy. Water is denser than ice, so the average molecular separation decreases on melting. Only statement (3) is correct. Answer: B.

Question 3 *

A weather balloon filled with 21 kg of helium gas has a volume of 120 m^3 at 27°C . Find the gas pressure in the balloon.

Given: mass of one mole of helium gas = $4 \times 10^{-3} \text{ kg}$

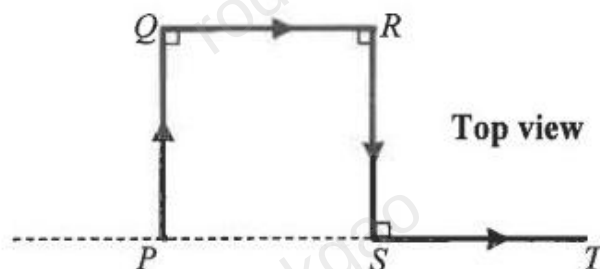
- A. $9.93 \times 10^4 \text{ Pa}$
- B. $1.00 \times 10^5 \text{ Pa}$
- C. $1.05 \times 10^5 \text{ Pa}$
- D. $1.09 \times 10^5 \text{ Pa}$

Official answer: D

The amount of helium is $n = 21/(4 \times 10^{-3}) = 5250 \text{ mol}$, and $T = 27 + 273 = 300 \text{ K}$. From $pV = nRT$, $p = 5250(8.31)(300)/120 = 1.09 \times 10^5 \text{ Pa}$. Answer: D.

Question 4

A car travels with constant speed v along a horizontal road PQRST with four sections of equal length as shown. Determine v if the car's average velocity for the journey PQRST is 20 km h^{-1} in magnitude.



- A. 10 km h^{-1}
- B. 20 km h^{-1}
- C. 40 km h^{-1}
- D. It cannot be found as the length of each section is unknown.

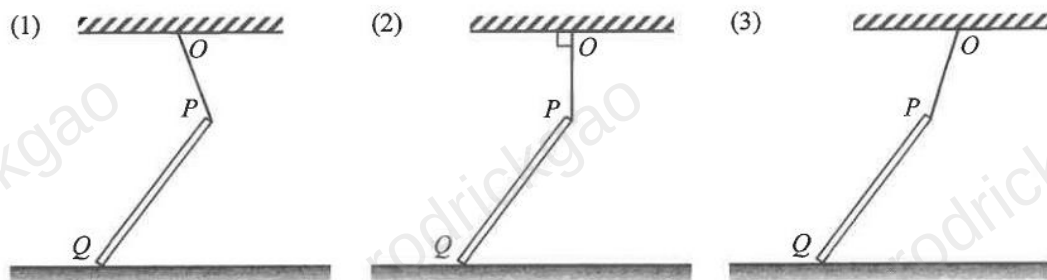
Official answer: C

Let each road section have length L . The distance travelled is $4L$, while the magnitude of displacement PT is $2L$. Hence the magnitude of average velocity is $(2L)/(4L/v) = v/2 = 20 \text{ km h}^{-1}$, giving $v = 40 \text{ km h}^{-1}$. Answer: C.

Question 5

In each figure below, a uniform rod PQ has its upper end P attached to point O on the ceiling by a light inextensible string. The rod's lower end Q rests on a rough horizontal ground as shown.

In which figure is the frictional force acting on the rod due to the ground towards the right?



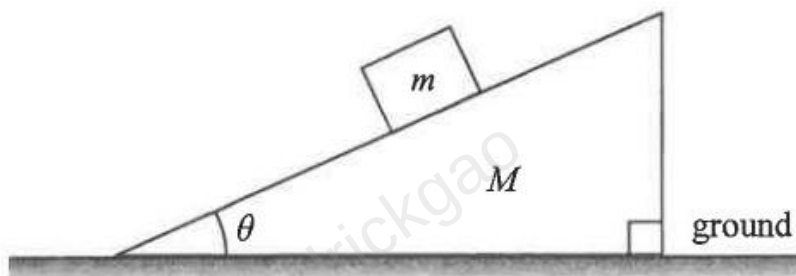
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: A

Tension acts from P towards O. In figure (1), its horizontal component is to the left, so friction must act to the right for horizontal equilibrium. In figure (2) the tension is vertical and no horizontal friction is needed. In figure (3), tension points up and right, so balancing friction would point left. Only figure (1) has the required rightward friction. Answer: A.

Question 6

A block of mass m is placed on a wedge of mass M as shown. The system remains at rest. What is the normal force exerted on the wedge by the ground?



- A. Mg
- B. $(M + m)g$
- C. $Mg + mg \cos \theta$
- D. $Mg + mg \tan \theta$

Official answer: B

Treat the block and wedge as one stationary system. The ground supports their combined weight, so its normal force is $(M+m)g$. Forces between the block and wedge are internal and do not change this total. Answer: B.

Question 7

An egg is released from a height and falls towards a cushion without breaking. Which of the following is the most probable explanation?



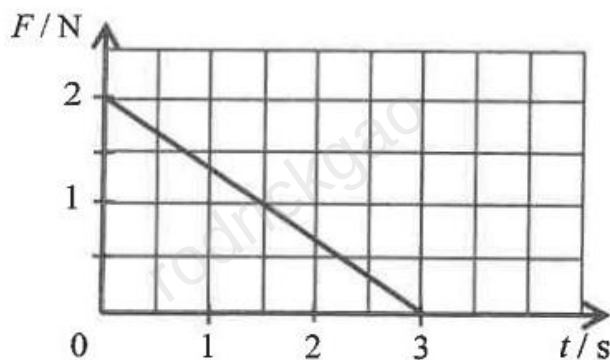
- A. The cushion makes the egg stop at a shorter distance.
- B. The cushion helps lengthen the time of impact.
- C. The cushion helps reduce part of the gravitational force acting on the egg.
- D. The cushion increases the reaction force acting on the egg during impact.

Official answer: B

The egg undergoes approximately the same momentum change while stopping. The cushion increases the stopping time, so the average resultant force, $\Delta p / \Delta t$, is smaller and the egg is less likely to break. Answer: B.

Question 8

An object of mass 2 kg moves with velocity 1 m s^{-1} initially. It is acted upon by a force F which varies with time as shown in the graph. The force and the object's velocity are in the same direction. Find the speed of the object at $t = 3 \text{ s}$.



A. 1.5 m s^{-1}

B. 2.5 m s^{-1}

C. 3.5 m s^{-1}

D. 4.5 m s^{-1}

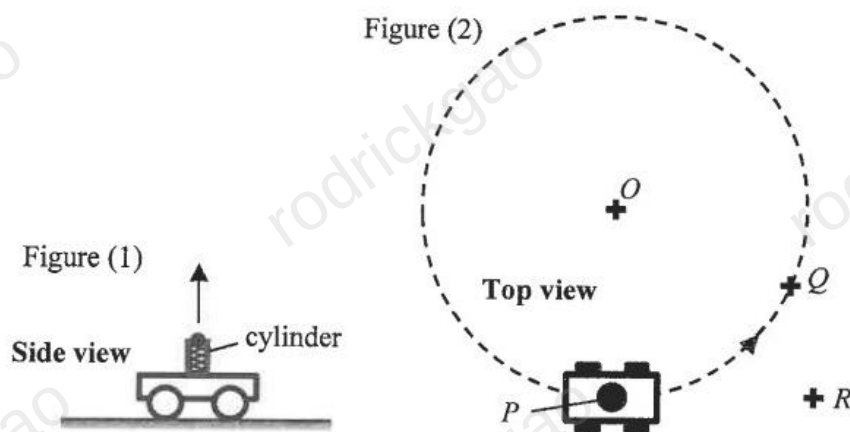
Official answer: B

The impulse is the area under the force–time graph: $J = \frac{1}{2}(3 \text{ s})(2 \text{ N}) = 3 \text{ N s}$. Using $J = m(v-u)$, $v = 1 + 3/2 = 2.5 \text{ m s}^{-1}$. Answer: B.

Question 9 *

A trolley with a spring-loaded launcher in Figure (1) can project a ball vertically upward. The trolley travels on the ground at a constant speed in a horizontal circle (with centre O) as shown in Figure (2).

The ball is projected when the trolley is at point P. After some time, the ball falls back to the ground as the trolley just reaches point Q. Neglecting air resistance, where would the ball land?



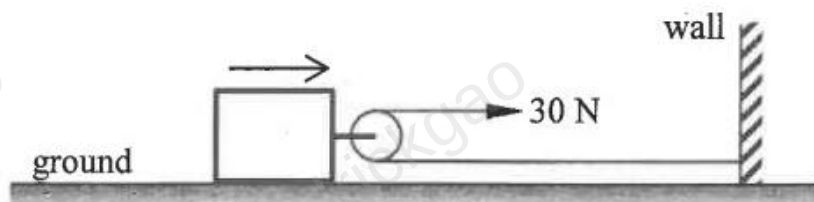
- A. O
- B. P
- C. Q
- D. R

Official answer: D

On release, the ball retains the trolley's instantaneous horizontal velocity, tangent to the circle at P. With no horizontal force, that velocity remains constant while the ball rises and falls. It therefore lands at R on the tangent, while the trolley turns towards Q. Answer: D.

Question 10

On a horizontal ground there is a block attached with a smooth light pulley as shown. A light and inextensible horizontal string, with one end fixed to a wall, passes the pulley. A man exerts a horizontal force of 30 N at the other end of the string and pulls a distance of 4 m. Find the work done by the man if the ground exerts a frictional force of 10 N on the block.



- A. 20 J
- B. 80 J
- C. 100 J
- D. 120 J

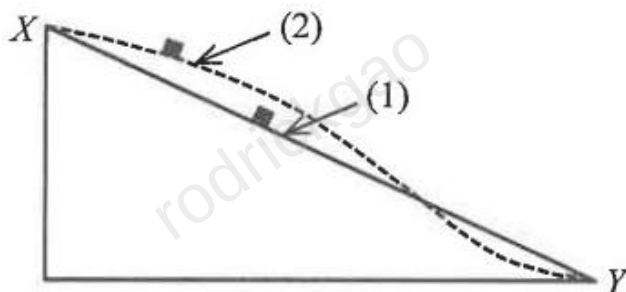
Official answer: D

Work done by the man is his force multiplied by the displacement of the end he pulls: $W = Fs = 30(4) = 120 \text{ J}$. The block moves only 2 m; 20 J is dissipated against friction and the remaining 100 J becomes kinetic energy. The question asks for the man's total work. Answer: D.

Question 11

In the figure below, the straight path (1) and the curved path (2) in a vertical plane are smooth. A small block slides from rest at X along the respective paths to Y.

Which of the following about the speed of the block at Y and the time taken to reach Y is correct? Neglect air resistance.



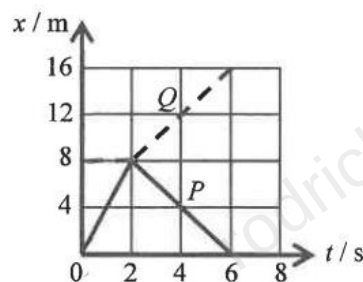
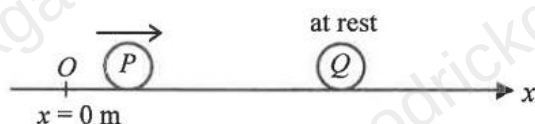
- A. the same speed; different time
- B. the same speed; the same time
- C. different speed; the same time
- D. different speed; different time

Official answer: A

Both smooth paths have the same vertical drop, so conservation of mechanical energy gives the same final speed, $v = \sqrt{2gh}$. Their slopes and acceleration histories differ, so their travel times differ. Answer: A.

Question 12

On a smooth horizontal surface, sphere P travelling along the x-axis collides head-on with another sphere Q, which is initially at rest at $x = 8$ m. The graph shows the displacement–time (x – t) relationship of P (solid line) and Q (dotted line). The collision takes place at $t = 2$ s and the collision time is negligible. Which statement is correct?



- A. The collision is perfectly inelastic.
- B. The mass of P is larger than that of Q.
- C. P still moves along +x direction after collision.
- D. P and Q move with the same speed after collision.

Official answer: D

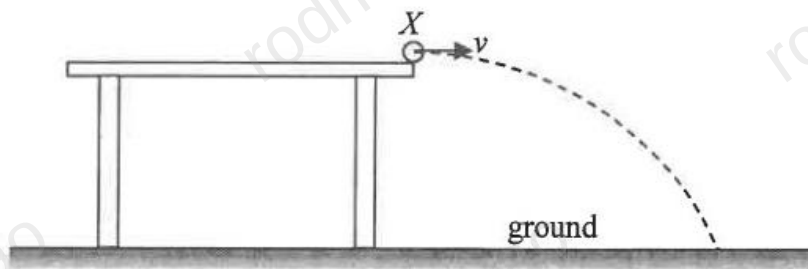
Velocity is the slope of the displacement–time graph. After impact, $v_P = (0-8)/(6-2) = -2 \text{ m s}^{-1}$ and $v_Q = (16-8)/(6-2) = +2 \text{ m s}^{-1}$. Thus their speeds are equal but their directions are opposite. They do not stick together. Answer: D.

Question 13 *

A marble leaves the edge of a table at point X with a horizontal speed v and hits the ground at a certain point as shown.

If the marble leaves the table at a higher speed, which of the following would remain unchanged? Neglect air resistance.

- (1) The time of flight of the marble in the air.
- (2) The acceleration of the marble during its flight in air.
- (3) The horizontal distance between X and the landing point.



- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: C

Vertical motion is independent of the initial horizontal velocity. For the same table height h , $t = \sqrt{2h/g}$ and the acceleration g downward are unchanged. The horizontal range vt increases. Statements (1) and (2) are correct.

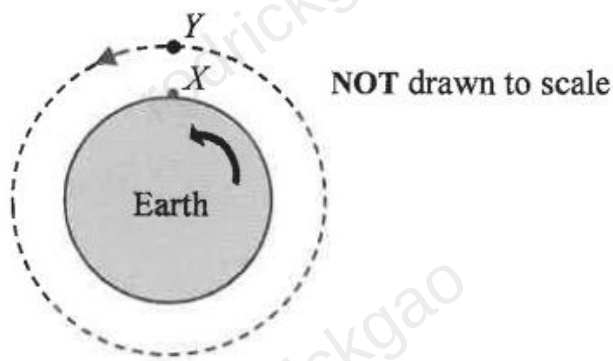
Answer: C.

Question 14 *

In the figure, X is an object resting on the Earth's equator. Y is a geostationary satellite moving in a circular orbit above the equator such that it always appears stationary to an observer on Earth.

Which descriptions about the motion of X and Y are correct?

- (1) The motion of X and Y takes the same period.
- (2) X is moving with a slower speed.
- (3) X has a greater acceleration.



- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

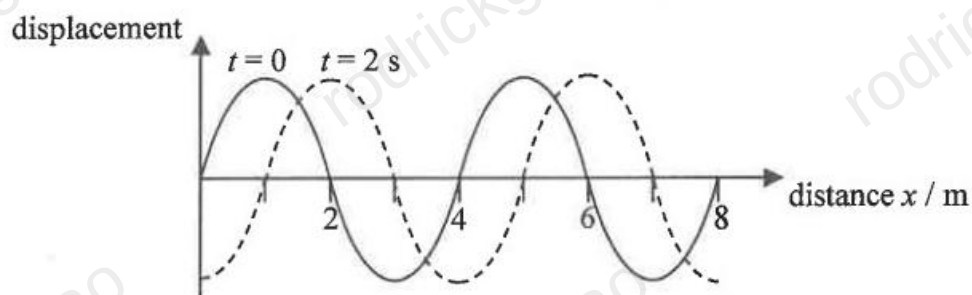
X and Y rotate with Earth and have the same period and angular speed. Since $v = \omega r$ and $a = \omega^2 r$, the smaller orbital radius of X gives both a smaller speed and a smaller centripetal acceleration. Statements (1) and (2) are correct. Answer: A.

Question 15

The figure shows the displacement–distance graph at time $t = 0$ and $t = 2$ s respectively of a wave travelling to the right.

Which statements about the wave are correct?

- (1) The wavelength is 4 m.
- (2) The period is 4 s.
- (3) The speed of the wave can be 2.5 m s^{-1} .



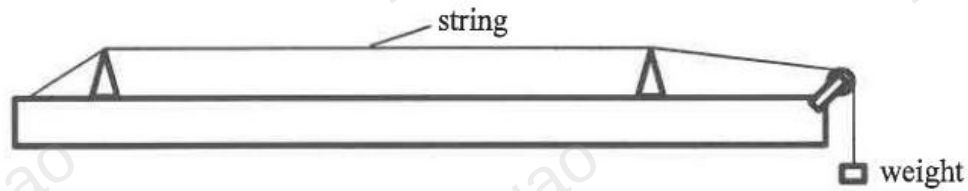
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: B

The wavelength read from successive crests is 4 m. The later profile is displaced rightward by 1 m plus any whole number of wavelengths, so $v = (1+4n)/2 \text{ m s}^{-1}$ for $n = 0, 1, 2, \dots$. In particular, $v = 2.5 \text{ m s}^{-1}$ is possible when $n = 1$. A period of 4 s would shift the profile by 2 m in 2 s, inconsistent with the graph. Statements (1) and (3) are correct. Answer: B.

Question 16

Referring to the set-up below, which of the following combinations would give the highest speed of wave propagation along the string when the string is plucked?



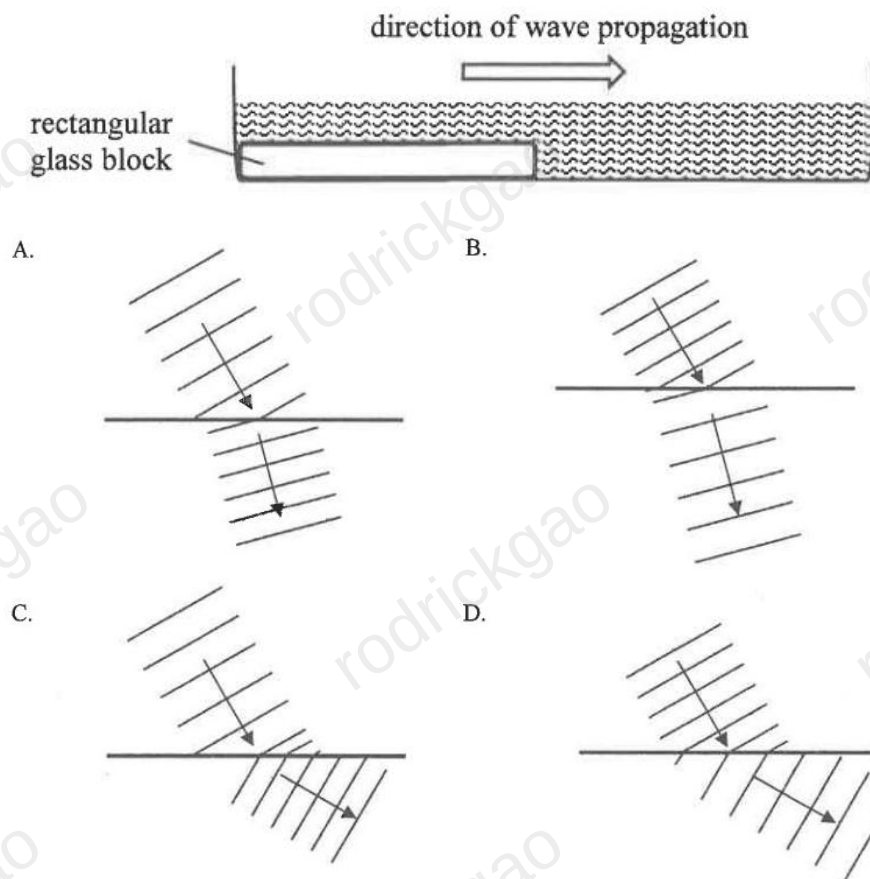
- A. tension T ; radius r
- B. tension T ; radius $2r$
- C. tension $2T$; radius r
- D. tension $2T$; radius $2r$

Official answer: C

For strings of the same material, $v = \sqrt{T/\mu}$ and $\mu = \rho\pi r^2$. Wave speed is greatest for the largest tension and smallest radius: tension $2T$ and radius r . Answer: C.

Question 17

Plane water waves travel from shallow water to deep water in a ripple tank as shown below. Which diagram correctly shows the top view of the wavefronts in the ripple tank?

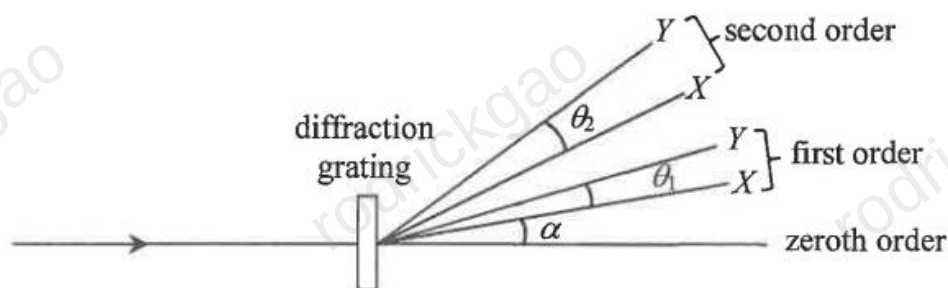


Official answer: D

Water waves speed up in deeper water. Their frequency remains unchanged, so their wavelength increases and the direction of propagation bends away from the normal. Diagram D shows both the larger wavefront spacing and the correct bending. Answer: D.

Question 18 *

A light beam which consists of monochromatic lights X and Y is incident normally on a diffraction grating. The figure shows the spectra of the first two orders. The diffraction angle of X in the first order is α . The angular separation between X and Y in the first order is θ_1 and that in the second order is θ_2 . Which statement must be correct?



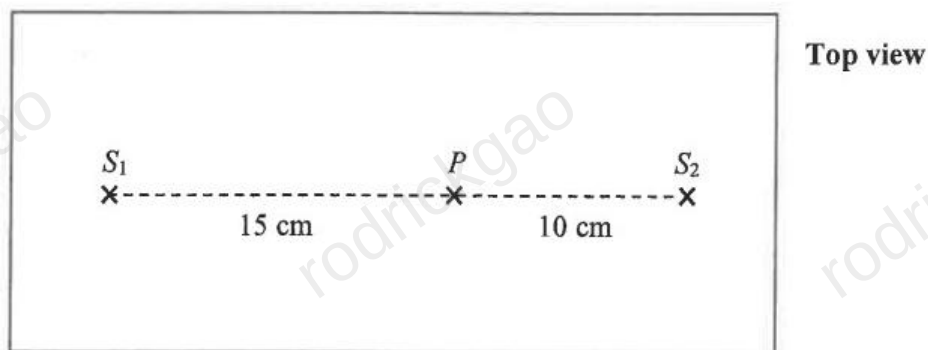
- A. The wavelength of Y is shorter than that of X.
- B. α would be larger if the grating spacing is smaller.
- C. θ_1 is independent of the grating spacing.
- D. $\theta_2 = 2\theta_1$.

Official answer: B

The grating relation is $d \sin \alpha = \lambda_X$ for first order. Decreasing d increases α . Y has the larger diffraction angle and hence the longer wavelength. The angular separation depends on d and the nonlinear sine relation, so the second-order separation is not exactly twice the first-order separation. Answer: B.

Question 19

In the ripple tank shown below, dippers S_1 and S_2 are vibrating in phase with frequency f and generate two sets of waves towards each other at a speed of 20 cm s^{-1} . Destructive interference occurs at point P on the straight line S_1S_2 . Which of the following is a possible value of f ?



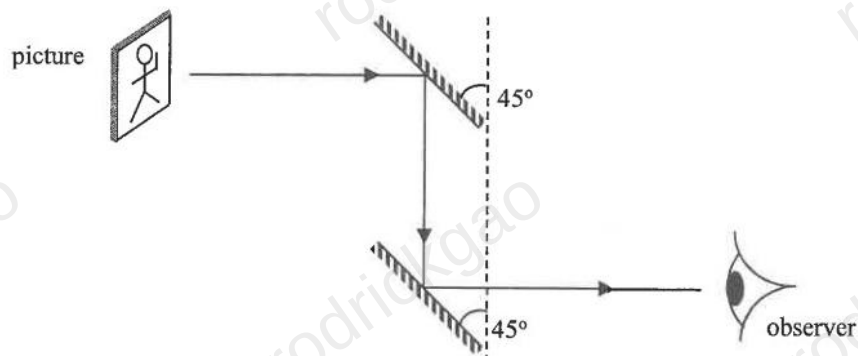
- A. 24 Hz
- B. 20 Hz
- C. 18 Hz
- D. 16 Hz

Official answer: C

The path difference at P is $15 - 10 = 5 \text{ cm}$. For in-phase sources, a minimum requires $\Delta = (n + \frac{1}{2})\lambda$, so $f = v(n + \frac{1}{2})/\Delta = 4(n + \frac{1}{2}) \text{ Hz}$. The allowed sequence is 2, 6, 10, 14, 18, ... Hz; among the options only 18 Hz is possible. Answer: C.

Question 20

A periscope is formed by two plane mirrors as shown. Which image will the observer see?



A.



B.



C.



D.



Official answer: A

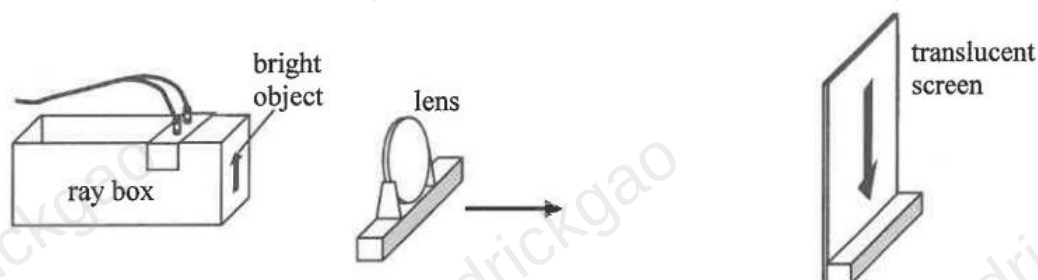
Each plane mirror reverses the component of the image perpendicular to its plane. The two parallel mirrors produce two reflections, restoring the original orientation and handedness. The observer sees the upright figure with its raised arm on the side shown in A. Answer: A.

Question 21

In the set-up below, the separation between the bright object and the translucent screen is fixed. A lens is moved from the object towards the screen as shown.

The first sharp image formed is inverted as shown and its length is 9 cm. When the lens is moved further towards the screen, a second sharp image of length 1 cm is observed. Which of the following statements is/are correct?

- (1) The second image is erect.
- (2) The length of the object is 3 cm.
- (3) There are at most two positions of the lens that can give a sharp image on the screen when the lens is moved.



- A. (1) only
- B. (2) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: C

For fixed object–screen separation, the two sharp-image positions interchange object and image distances. Their magnifications are reciprocal, so $h_1 h_2 = h_{\text{object}}^2$ and $h_{\text{object}} = \sqrt{(9 \times 1)} = 3$ cm. Both screen images are real and inverted. The lens equation gives at most two positions. Statements (2) and (3) are correct. Answer: C.

Question 22

Which statements about ultrasound are correct?

- (1) Ultrasound is longitudinal waves.
- (2) Ultrasound needs media to propagate.
- (3) The speed of ultrasound in glass is higher than that in air.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

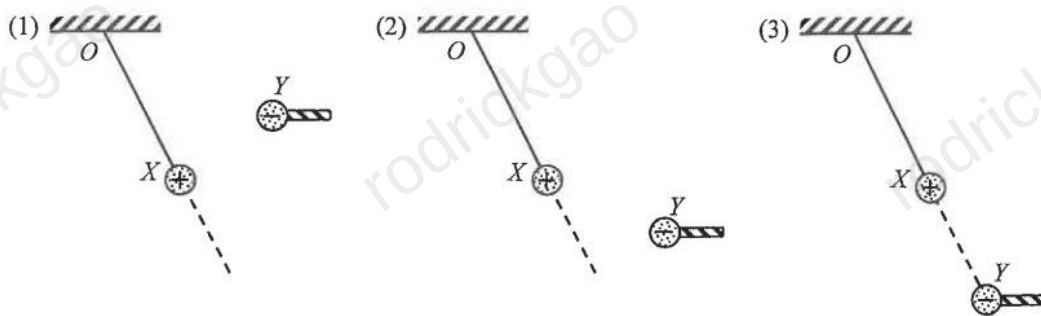
Official answer: D

Ultrasound is a mechanical longitudinal wave in the media considered here and requires a medium. Its propagation speed depends on the medium and is greater in glass than in air. All three statements are correct. Answer: D.

Question 23

A positively charged sphere X of mass m is suspended from a fixed point O by a nylon thread. Another negatively charged sphere Y at the end of an insulating rod is held in various positions as shown. O, X and Y are in the same vertical plane.

In which situation(s) could X be kept at rest as shown?



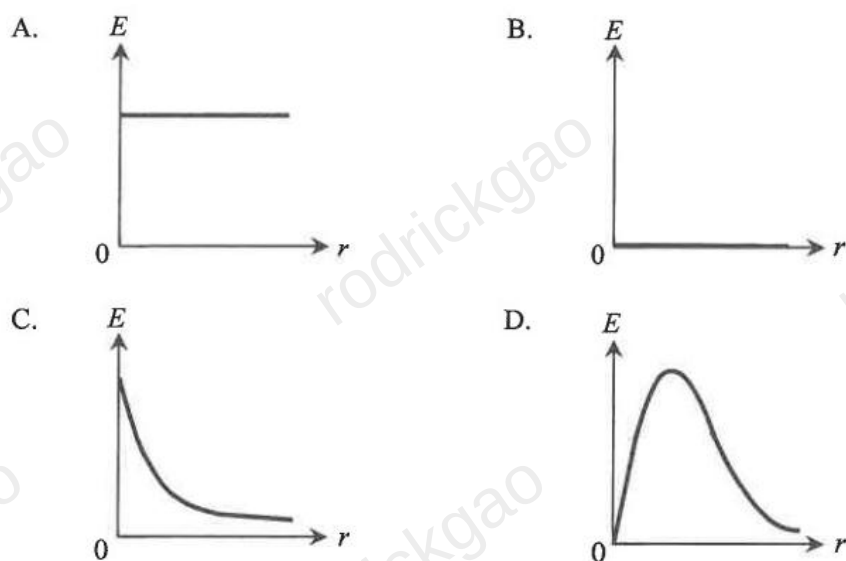
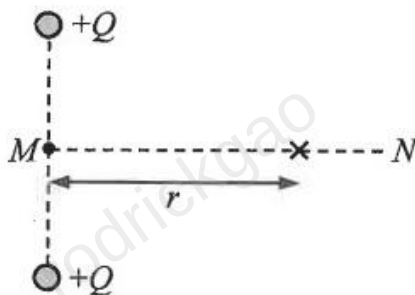
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: C

The attractive electric force on X points towards Y, while tension points towards O and weight acts downwards. In figures (1) and (2), the electric force can provide the anticlockwise moment that balances the weight's clockwise moment about O. In figure (3), the electric force and tension lie along OX and provide no moment about O, so the weight cannot be balanced. Answer: C.

Question 24 *

Two positive point charges $+Q$ are fixed as shown. MN is the perpendicular bisector of the line joining the charges. Which graph correctly shows the variation of the electric field strength on line MN with distance r from M?



Official answer: D

At midpoint M the two fields cancel. At a point a distance r along MN, their transverse components cancel and their components along MN add: $E = 2kQr/(r^2+a^2)^{3/2}$, where $2a$ is the charge separation. Thus E starts at zero, rises to a maximum and tends to zero as r becomes large. Answer: D.

Question 25

An electrical device is charged with a steady d.c. of 60 mA for 30 minutes. Find the number of electrons that pass through the device during charging.

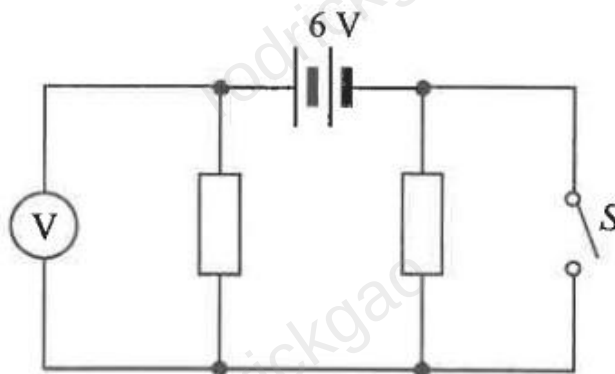
- A. 108
- B. 1800
- C. 3.75×10^{17}
- D. 6.75×10^{20}

Official answer: D

The transferred charge is $Q = It = (60 \times 10^{-3})(30 \times 60) = 108 \text{ C}$. The number of electrons is $N = Q/e = 108/(1.60 \times 10^{-19}) = 6.75 \times 10^{20}$. Answer: D.

Question 26

In the above circuit, the resistors are identical and the 6 V battery has negligible internal resistance. Which of the following is the set of readings on the voltmeter when (1) S is open; (2) S is closed?



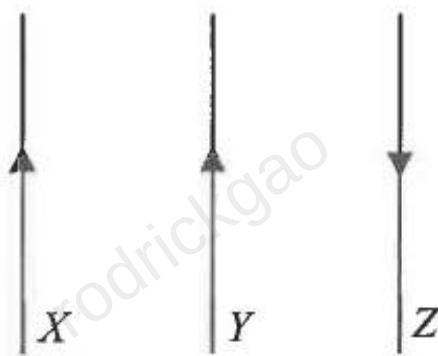
- A. S open: 0 V; S closed: 6 V
- B. S open: 3 V; S closed: 6 V
- C. S open: 6 V; S closed: 0 V
- D. S open: 6 V; S closed: 3 V

Official answer: B

With S open, the identical resistors are in series and share the 6 V supply equally, so the voltmeter reads 3 V. Closing S short-circuits the right-hand resistor, leaving the left-hand resistor across the full supply: 6 V. Answer: B.

Question 27

The figure below shows three parallel wires carrying currents in the directions shown. If one of the wires experiences zero resultant magnetic force, that wire



- A. must be X.
- B. must be Y.
- C. must be Z.
- D. may be Y or Z.

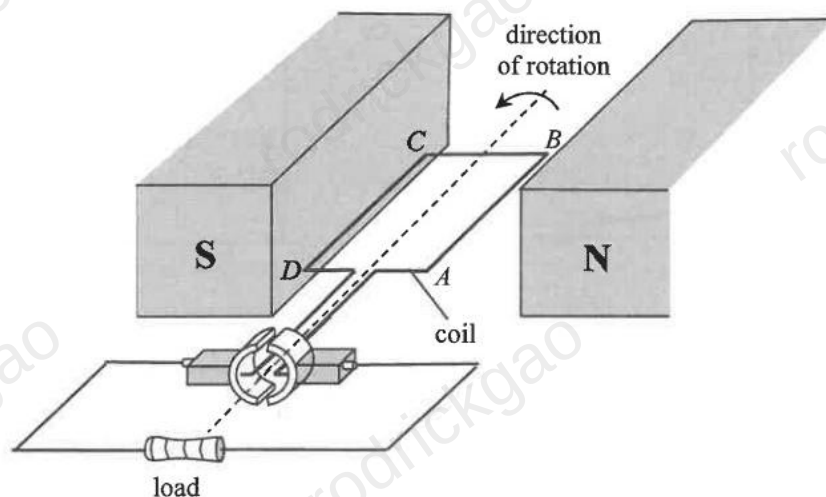
Official answer: A

Parallel currents attract and antiparallel currents repel. On Y, attraction towards X and repulsion from Z both act leftward. On Z, repulsion from X and Y both act rightward. Only X experiences opposing forces, whose magnitudes can balance. Answer: A.

Question 28

The figure shows the structure of a simple generator. Which of the following statements is/are correct?

- (1) The magnetic force acting on side AB of the coil is upward at the instant shown.
- (2) The commutator would reverse the direction of current in the coil whenever the coil passes its vertical position.
- (3) The current flowing through the load is an unsteady d.c.



- A. (1) only
- B. (2) only
- C. (3) only
- D. (1) and (3) only

Official answer: C

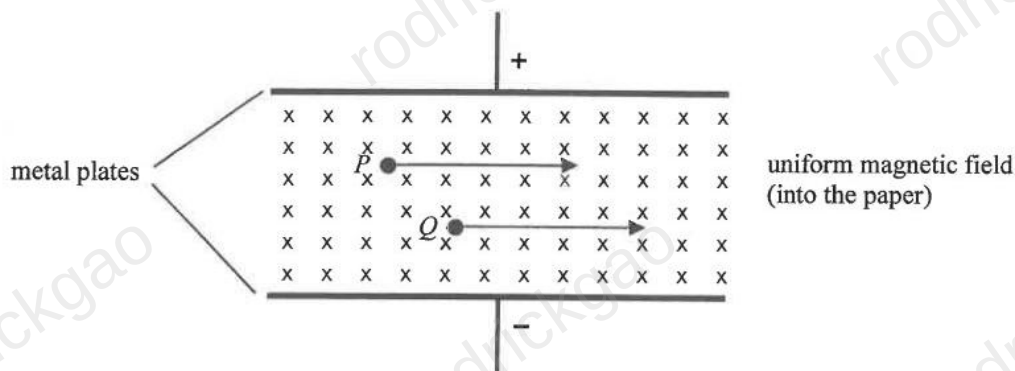
The induced current produces a magnetic force opposing the rotation; on AB it is downward at the instant shown. The induced e.m.f. in the rotating coil reverses because the motion relative to the field reverses, while the split-ring commutator swaps the external connections each half-turn. The load therefore receives a pulsating, unidirectional current. Only statement (3) is correct. Answer: C.

Question 29 *

Charged particles P and Q are moving in a region with mutually perpendicular uniform electric and magnetic fields as shown.

If both particles are not deflected by the fields, which of the following statements must be correct? Neglect the effects of gravity.

- (1) They are positively charged.
- (2) They are moving with the same velocity.
- (3) They have the same charge-to-mass ratio.



- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: B

For no deflection, the electric and magnetic forces cancel: $|q|E = |q|vB$, giving $v = E/B$. Both particles move rightward as shown and have the same speed, so their velocities are equal. Reversing the charge reverses both forces and still permits cancellation; neither the charge sign nor charge-to-mass ratio is fixed. Answer: B.

Question 30 *

Arrange the following in ascending order of power delivered when each of them is connected to the same resistor.

- (1) a 100 Hz sinusoidal a.c. of peak voltage 2 V
- (2) a 50 Hz sinusoidal a.c. of r.m.s. voltage 2 V
- (3) a steady d.c. of voltage 1.5 V

A. (1), (3), (2)

B. (2), (3), (1)

C. (1), (2), (3)

D. (2), (1), (3)

Official answer: A

Mean power is V_{rms}^2/R . Case (1) gives $(2/\sqrt{2})^2/R = 2/R$; case (2) gives $4/R$; case (3) gives $1.5^2/R = 2.25/R$. Hence the ascending order is (1), (3), (2). Frequency does not change these mean powers for the same ideal resistor. Answer: A.

Question 31

Which of the following statements about X-rays is INCORRECT?

A. X-rays can be produced by high-speed electrons hitting a metal target.

B. X-rays are a kind of electromagnetic waves.

C. X-rays can be deflected by an electric field.

D. Although X-rays do not carry charges, they can cause ionization.

Official answer: C

X-rays are uncharged electromagnetic radiation and are not deflected by an electric field. They can nevertheless ionize matter by transferring photon energy to electrons. Statement C is the incorrect statement. Answer: C.

Question 32 *

Radioactive sources X and Y have initial activities 100 kBq and 200 kBq respectively. One day later, their activities are found to be equal. After another day, the activity of X becomes 80 kBq. Deduce the corresponding activity of Y.

- A. 40 kBq
- B. 50 kBq
- C. 89 kBq
- D. 160 kBq

Official answer: A

Let the one-day activity factors be a for X and b for Y. Equality after one day gives $100a = 200b$, hence $a = 2b$. After two days, $100a^2 = 80$. Therefore $A_Y = 200b^2 = 200a^2/4 = 40$ kBq. Answer: A.

Question 33

What is the value of x for the fission reaction below?



- A. 1
- B. 2
- C. 3
- D. 4

Official answer: C

Conserve nucleon number: $235+1 = 92+141+x$, so $x = 3$. Proton number also balances: $92 = 36+56$. Answer: C.